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Ergogenic Aids in Sport

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In Closing

In the summer of 2013, two world-class track sprinters failed their in-competition urine tests and were provisionally suspended until each of their urine B samples was analyzed in the presence of the athletes or their representatives. The B-sample testing confirmed the results of the A-sample testing: The athletes had ingested oxilofrine, a banned stimulant. Each sprinter faced a suspension of up to 4 years, a ban that would effectively end their competitive careers. Both sprinters professed their innocence, claiming that they must have consumed a dietary supplement that was contaminated with the drug. Subsequent analysis of one of the many supplements the athletes admitted consuming showed that the supplement did indeed contain oxilofrine. Athletes are held fully responsible—strictly liable—for everything they ingest, so even when athletes inadvertently ingest a banned substance, ignorance of supplement contamination is not a valid defense. Unfortunately for the athletes, it is entirely possible that they had no prior knowledge that the supplement they willingly ingested on the advice of a personal trainer contained a banned substance.

Some dietary supplements purported to improve muscle strength, sexual performance, or weight loss have been found to contain banned substances—often prescription medications or altered versions thereof. Surprising to many people is that some supplements even have banned substances listed as ingredients on the product labels. Setting aside the accidental contamination of supplements, the never-ending quest to improve performance, lose weight, or alter body image has led some athletes to purposefully use banned substances or techniques. Decades of purposeful attempts at cheating have led to the implementation of sophisticated drug testing administered by the World Anti-Doping Agency (WADA) and other sports organizations to catch athletes who try to gain an unfair advantage over their competitors. Unfortunately, some athletes fail sport drug tests because they ingested supplements that, unbeknownst to them, contained prohibited ingredients. Even though they had no intent to cheat, those athletes also risk long suspensions from their sports, large legal fees, and loss of their good reputations.

Some athletes are willing to try anything to improve their performance and believe that special dietary supplements can be the deciding factor. Others choose to use prohibited techniques such as blood doping. Still others try prescription medications or various formulations of hormones in an attempt to create a competitive advantage.

Ergogenic aids are substances or practices (such as hypnosis or a coach's pep talk) that improve an athlete's performance. Some

ergogenic aids, such as proper training and basic sport nutrition, are a legitimate part of sport, and their benefits have been extensively researched. However, the proposed benefits of many **ergogenic** (work-producing) substances or techniques lack the scientific support required to verify their effectiveness. Athletes often receive tips about ergogenic aids from friends or coaches or have seen advice on the Internet and assume that such guidance is safe and accurate. Some athletes experiment with dietary supplements, special clothing and equipment, or unique training techniques hoping for even a slight performance improvement, regardless of possible negative consequences.

The list of ergogenic aids is long, but the list of those with substantiated ergogenic properties is much shorter. In fact, some allegedly ergogenic substances or practices can actually impair performance. These are usually drugs—beta-blockers are one example—and one author has termed them **ergolytic** (work-decreasing) agents.²² Ironically and sometimes tragically, several ergolytic agents have been promoted as ergogenic aids.

Many athletes indiscriminately take prescription drugs and dietary supplements in the belief that these will improve their performance. According to one study of 53 Division I university coaches and trainers, 94% provided their athletes with dietary supplements, despite the fact that the National Collegiate Athletic Association (NCAA) and the National Athletic Trainers' Association (NATA) encourage nutrition education and a food-first approach rather than reliance on dietary supplements.^{14,47} Use of dietary supplements might seem like a legitimate attempt to improve performance, but some dietary supplements are purposefully or accidentally contaminated with prohibited substances.

The focus of this chapter is on selected nutritional supplements with possible ergogenic effects, as well as banned substances and techniques with confirmed ergogenic effects. More general sport nutrition practices are addressed in [chapter 15](#). Psychological practices and mechanical factors are beyond the scope of this book but are reviewed in depth elsewhere.⁵⁶

Before we cover specific ergogenic aids, it is important to take a quick look at some of the challenges associated with studying the

proposed effects of performance-enhancing agents. For example, the inherent limitations of designing studies on ergogenic aids, including the power of the placebo effect, directly affect our ability to draw practical and scientific conclusions. For those reasons, many well-designed studies are usually required before it is possible to confidently draw conclusions about the ergogenic benefit of any intervention.

Researching Ergogenic Aids

Anyone can claim that a certain substance or technique is ergogenic, but before an intervention can be legitimately classified as truly performance enhancing, it must be shown by competent research to consistently enhance performance in a variety of relevant circumstances. High-quality scientific studies are essential to differentiate between a true ergogenic response and a placebo response, in which performance improves simply because research subjects (or athletes) expect improvement.

The Placebo Effect

Assume that an athlete consumes a dietary supplement several hours before game time and then has a successful performance. The athlete might attribute the success to the supplement, even though there is no evidence that ingesting the supplement could have had any effect whatsoever. This is a good example of why anecdotal accounts are not a substitute for valid and reliable scientific data.

As discussed in the introductory chapter, the phenomenon by which one's expectations affect the body's response in some measurable way is known as the **placebo effect**. From a practical standpoint, a placebo effect can improve performance simply because the athlete expects an improvement. For that reason, placebo effects can be beneficial. For example, if a coach tells a young athlete that his or her new shoes will definitely help with running faster, there is a good chance the athlete will run faster. The placebo effect can be helpful from a practical perspective, but from a scientific perspective, the placebo effect can complicate studies of ergogenic aids because researchers must be able to separate the placebo effect from the true physiological responses to the intervention being tested.

The power of the placebo effect was clearly demonstrated in one of the earliest studies of **anabolic steroids** in sport.⁴ Fifteen male athletes who had been involved in heavy weightlifting for the previous 2 years volunteered for a strength training experiment using anabolic steroids. They were told that those who made the greatest strength gains over a preliminary 4-month weight training period would be selected for the second phase of the study, in which they would receive anabolic steroids.

Following the initial period, eight of these 15 subjects were randomly selected to enter the treatment phase. Only six of these subjects passed all medical screening tests and were allowed to continue the study. The next phase of the study consisted of a 4-week period in which the subjects were told that they would receive 10 mg per day of Dianabol (a potent anabolic steroid), when in fact they received a placebo—an inactive substance typically provided in a form that looks identical to the genuine drug.

Strength data were collected over the last 7 weeks of the 4-month pretreatment training period and over all 4 weeks of the treatment (placebo) period (see [figure 16.1](#)). Even though the subjects were experienced weightlifters, they continued to gain impressive amounts of strength during the pretreatment training period. However, strength gains while subjects were taking the placebo were even greater than during the pretreatment period! The group improved by an average of 11 kg (24 lb) in combined 1-repetition maximum (1RM) lift totals (squat, bench press, military press, sitting press) during the 7-week pretreatment period but improved 45 kg (~100 lb) during the 4-week treatment (placebo) period. This represents an average gain in strength of 1.6 kg (3.5 lb) per week during the pretreatment training period and 11.3 kg (25 lb) per week during the placebo period—an increase in the rate of strength gain more than seven times greater during the placebo (supposed steroid) period over the pretreatment training period.

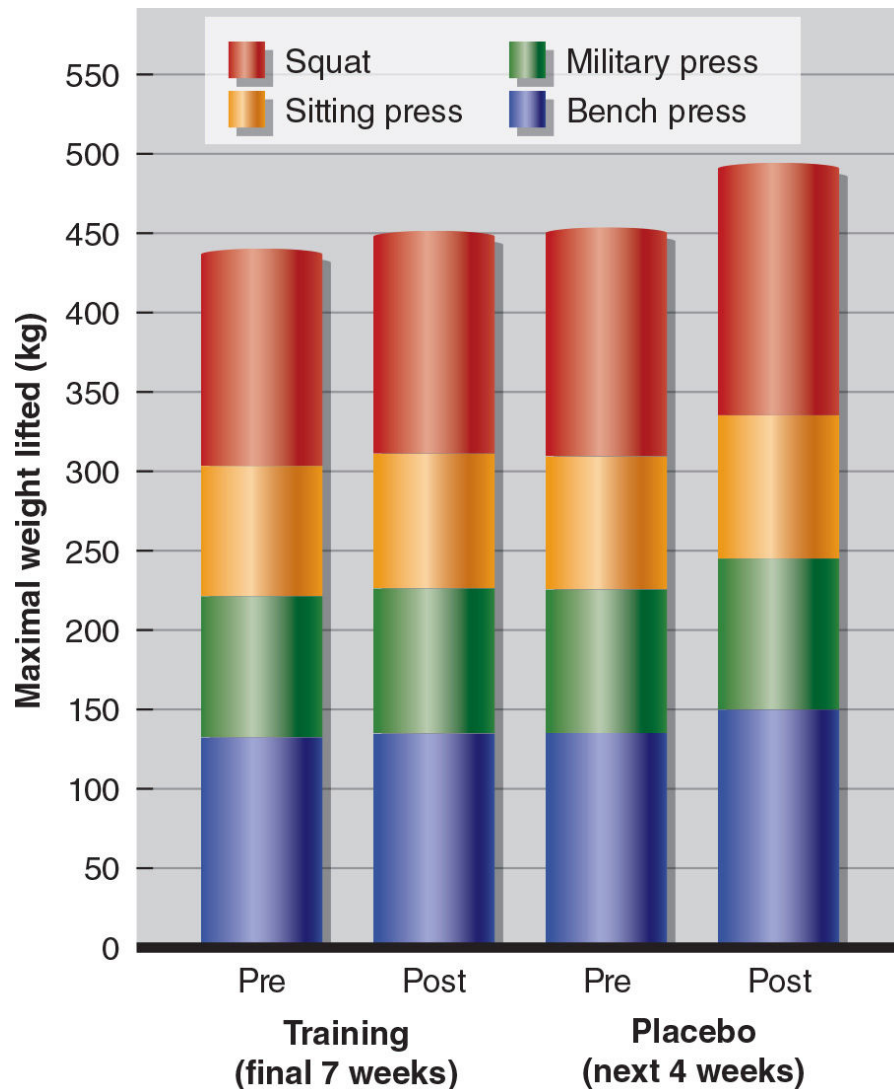


FIGURE 16.1 The placebo effect on muscular strength gains. The increase in total strength and strength in each of four maximum lifts over the last 7 weeks of an intense 4-month pretreatment training period is compared with strength increases during a subsequent 4-week treatment period, in which the subjects took placebos that they thought were anabolic steroids and continued intense resistance training.

Data from Ariel and Saville (1972).



From a practical standpoint, a placebo effect can certainly be a good thing for sport performance, provided that the related risks are acceptable. For example, football players may breathe from an

oxygen tank on the sidelines, but there is no scientific evidence that breathing oxygen-enriched air will speed recovery during football games. But if athletes believe that they will recover faster, that very belief is likely to make them feel better. Such is the power of the placebo effect, an often effective approach to reducing feelings of pain and fatigue.

RESEARCH PERSPECTIVE 16.1

The “Nocebo” Effect on Sport Performance

A placebo effect is defined as a positive psychobiological response to a purported beneficial treatment. A “nocebo” effect, conversely, is a negative psychobiological response to a purported harmful treatment, for example, when a patient is told to anticipate a side effect of a medication or that a certain exercise might cause pain. There is empirical support for both placebo and nocebo effects in sport performance. However, those studies have generally been too small to reliably identify the variables associated with the placebo and nocebo responses.

A recent study sought to estimate the relative magnitude of placebo and nocebo effects on sport performance.³⁴ Approximately 600 adults participated and were randomized to three different treatments: positive-belief treatment (participants were deceptively administered an inert capsule described as a potent supplement to improve spring performance), negative-belief treatment (participants were deceptively administered an inert capsule described as a potent supplement to negatively affect spring performance), or no-treatment control (participants received no instruction or capsule). Participants completed an exercise performance test consisting of five 20 m sprints before and after treatment.

In contrast to previous studies, there was no improvement in sprint performance following the ingestion of a purportedly beneficial supplement. However, the authors did observe a significant worsening in sprint performance in those individuals receiving a purportedly harmful supplement (i.e., a nocebo effect). Second, within the positive-belief treatment, the performance of participants reporting that they were likely to use a supplement was improved to a greater degree than in those participants not intending to use a supplement. In other words, the intention to use sports supplements might relate to placebo responding. This has practical implications for sports practitioners in that the effectiveness of a given treatment may relate to the athlete’s intentions to use it.

Rawdon and colleagues⁴² conducted a meta-analysis of 37 studies and confirmed the presence of a strong placebo effect associated

with sport supplements that claimed to improve performance. This finding may not be surprising, but it does confirm that the mere anticipation of a benefit from a dietary supplement is often associated with a benefit. When subjects received either the test supplement or a placebo, measures of muscular performance improved by a similar amount when compared to values in the control groups. The finding that the supplement treatments did not differ from the placebo treatments lends further support to the authors' conclusion that "any improvements seen with supplementation should not be attributed solely to the supplement, as it is acting in synergy with a training stimulus".

Controlling for the Placebo Effect

All studies of potential ergogenic substances must include a placebo group so that researchers can compare responses resulting from the test substance with those resulting from a placebo. In many studies, a *double blind* experimental design is used, in which neither the subject nor the experimenter knows who is getting the proposed ergogenic aid and who is getting the placebo. This is done to eliminate experimenter bias, whereby the experimenter's beliefs might affect the outcome of the study. In this design, the substances are assigned a code, and only an independent person not associated with the project has access to the codes until all data have been collected and analyzed.

Placebos have such a potentially powerful effect that sport scientists often recommend that studies of ergogenic aids include not only a placebo treatment (or group) but also a control treatment to help identify (control for) the placebo effect. For example, in a study designed to determine if consuming oak leaf extract improves muscle strength after 8 weeks of training, at least three experimental treatments would be needed: (1) oak leaf extract pill, (2) a placebo pill, and (3) no pill. If strength gains after training were similar for all three treatments, the researchers would conclude that oak leaf extract has no ergogenic effect. If both the oak leaf extract treatment and the placebo treatment were associated with greater strength gains than the control treatment, the conclusion would be that the gain in strength was caused by a placebo effect. If strength gains on the oak leaf extract treatment were significantly greater than on both

the placebo and the control treatments, only then could the researchers conclude that there was something in oak leaf extract that improves adaptations to strength training. In studying ergogenic aids, the addition of a control group improves the ability to understand the possible impact of the placebo effect.

Limitations of Ergogenic Aid Research

To establish the effectiveness (efficacy) of a possible ergogenic aid, scientists rely on accepted research methods to control for the placebo effect. However, even the most rigorously conducted studies can miss small effects that might actually be meaningful to athletes, whose success is often defined in fractions of a second. Such small differences in performance can sometimes be obscured by the statistical analyses that research requires.

Research can also be limited by the accuracy of lab equipment and the variability of research techniques. All research methods have some margin of error as well as inherent variability, and if the results fall within that margin of error, the researcher will be unable to isolate the effect of the intervention being tested. Because of measurement error, equipment variability, differences in individual responses, and the day-to-day variability of subjects' responses, a potential ergogenic aid must exert a major effect before scientific studies can confirm that it is ergogenic.

It is also important to keep in mind that performance in a laboratory is considerably different from performance in the athletic environment, so laboratory results do not always accurately reflect natural athletic results. Yet an advantage of laboratory testing is that the test environment can be carefully controlled. This is usually not possible in field studies, where several uncontrollable variables—such as temperature, humidity, wind, and distractions—can affect the results.

The ergogenic potential of any intervention cannot be confirmed by one study or, for that matter, even by a dozen studies. Dozens and sometimes hundreds of studies are required to draw a confident conclusion about whether or not an intervention is ergogenic. The ergogenic effect of sport drinks is a good example of the need for multiple studies to confirm efficacy. Before the late 1980s, the prevailing opinion among sport scientists was that sport drinks did not provide hydration or performance benefits differing from those seen

with plain water. This opinion was based on the scientific literature at the time, which contained little evidence that sport drinks were ergogenic. It was only after the publication of dozens of competent research studies conducted over many years that scientific opinion changed.

Another challenge in studying the potential benefits of dietary supplements is that researchers often have to assume that the contents of the supplement are accurately portrayed on the product label. This may not be the case. The values for active ingredients in the supplement may actually be considerably more or less than the values listed on the product label, or the supplement may be contaminated with a banned substance that could be responsible for whatever effect might be found. For this reason, some sport scientists argue that studies on dietary supplements should include third-party verification of the purity and potency of the supplement being tested.

RESEARCH PERSPECTIVE 16.2

Analgesic Use in Sport

For many years, clinicians have debated the use, misuse, and abuse of analgesics—painkillers like ibuprofen, naproxen, and aspirin—in sport. In 1967, the original International Olympic Committee prohibited narcotic analgesics from sport. The 2017 World Anti-Doping Agency prohibited list bans narcotics and cannabinoids. More commonly used analgesics, including nonsteroidal anti-inflammatory drugs, local anesthetics, and some weak opioids, however, are *not* prohibited. This leaves a gray area because no clear definition separates narcotics and cannabinoids from more commonly used analgesics in terms of their health risks or ergogenic potential.

This leads to one obvious question: What are the key determinants for including a substance on the prohibited list? The current criteria for the prohibition of a substance require consideration of the potential for, or enhancement of, sport performance, as well as the potential for, or actual, health risk. Whether use constitutes a violation of the spirit of sport (that is, the concept of fair, ethical, and respectful competition) is also considered. However, the specific health risks of narcotics and cannabinoids relative to other analgesics have not been rigorously assessed. In fact, there is even an argument that *all* opioids and cannabinoids should be at the physician's discretion in treating an athlete, but, currently, the window for using these substances in elite athletes is extremely narrow.⁵³

There are often passionate views of the role that the anti-doping code should play in regulating opioids and cannabinoids. Some suggest that the

social use of drugs might be better addressed by a code of conduct approach in which substance abuse counseling and treatment strategies would be employed. Others believe that the use of opioids and cannabinoids is a completely unacceptable behavior in the sport setting and potentially so dangerous that every effort should be taken to prevent their use; thus, drug testing and anti-doping sanctioning should continue. Moving forward, any approach to quell the debate surrounding the aforementioned gray area must be simultaneously grounded in the principles and best practices guiding pain management and treatment strategies in the sport setting.

Realizing that science progresses slowly, does so in small steps, and is limited in its ability to unequivocally determine the efficacy of performance-related interventions, we now turn our attention to examples of ergogenic aids.

Ergogenic Nutrition Aids

We continue this chapter with an overview of a few selected nutrition interventions that have been studied for their ergogenic effects, and we end the chapter by reviewing prohibited substances and techniques. Although basic concepts of nutrition and the specific performance-related properties of carbohydrate, fat, protein, vitamins, and minerals are discussed in [chapter 15](#), many nutrition interventions have been proposed to have specific ergogenic properties. We intentionally provide only a brief overview of a handful of nutrition interventions and encourage interested readers to seek additional information from the references provided. This section on ergogenic nutrition aids also highlights the necessity for dozens of well-designed research studies to be conducted on a particular topic before it is possible to confidently draw scientific conclusions and make practical recommendations.

It is impossible to cover all ergogenic nutrition aids being used by athletes, so the information in this section of the chapter is limited to the following:

- Bicarbonate
- β -alanine
- Leucine
- Caffeine

- Cherry juice polyphenols
- Creatine
- Nitrate

Bicarbonate

Recall from [chapter 7](#) that bicarbonate is an important part of the buffering system necessary to maintain the acid–base balance of body fluids. In highly anaerobic events, in which large amounts of lactic acid are formed, enhancing the body's buffering capacity through elevation of the blood's bicarbonate concentrations could improve performance capacity.

Proposed Ergogenic Benefits of Bicarbonate

Through ingestion of sodium bicarbonate (baking soda) or sodium citrate, the buffering capacity for metabolic acids such as lactic acid can be increased, allowing extracellular pH to be maintained even as active muscles release greater amounts of lactic acid. Theoretically, this could allow high-intensity exercise to continue for a longer time before fatigue sets in. Not surprisingly, oral intake of sufficient amounts of sodium bicarbonate does elevate plasma bicarbonate concentration. However, this has little effect on intracellular concentrations of bicarbonate. Therefore, the potential benefits of bicarbonate ingestion would theoretically be limited to anaerobic bouts of exercise lasting longer than 2 min, because bouts less than 2 min would be too brief to allow enough hydrogen ions (H^+ , from the lactic acid) to diffuse out of the muscle fibers into the extracellular fluid, where they could be buffered by the extra bicarbonate ions.

Demonstrated Effects of Bicarbonate

In 1990, Roth and Brooks⁴⁴ described a cell membrane lactate transporter that operates in response to the pH gradient between extracellular fluid and muscle cells. Increasing the extracellular buffering capacity by ingesting bicarbonate or citrate increases the extracellular pH, which in turn increases transport of lactate from the muscle fiber via the membrane transporter.

The theory underlying bicarbonate or citrate ingestion as a way to increase buffering capacity and improve high-intensity performance is sound, although the existing research literature is conflicting, in part

because of the relatively limited number of published studies. Linderman and Fahey,³⁸ in their review of the literature, found several important patterns in the research that might explain these conflicts. They concluded that bicarbonate ingestion had little or no effect on performances less than 1 min or more than 7 min in duration but that for performances between 1 and 7 min, the ergogenic effects were evident. Furthermore, they found that the dose was important. Most studies that used a dose of 300 mg/kg of body mass showed a benefit, whereas most studies using a lower dosage showed little or no benefit. Thus, it appears that bicarbonate ingestion of 300 mg/kg of body mass can enhance the performance of all-out, maximal anaerobic activities of 1 to 7 min duration. (Preliminary research indicates a possible synergistic effect of combining β -alanine and sodium bicarbonate,⁴⁶ although this possibility must be confirmed by additional research.)

An example of a study supporting this conclusion is illustrated in [figure 16.2](#). In this study, blood bicarbonate concentrations were artificially elevated by bicarbonate ingestion before and during five sprint-cycling bouts, each lasting 1 min (see [figure 16.2a](#)).²⁰ Performance on the final trial improved by 42%. This elevation in blood bicarbonate reduced the concentration of free H^+ both during and after exercise (see [figure 16.2b](#)), thereby elevating blood pH (keep in mind that pH is the negative logarithm of the H^+ concentration, so when H^+ falls, pH increases). The authors concluded that in addition to improving buffering capacity, the extra bicarbonate appeared to speed the removal of H^+ ions from the muscle fibers, thereby blunting the decrease in intracellular pH. This conclusion essentially predicted the presence of a lactate transporter in the muscle cell membrane that Roth and Brooks⁴⁴ reported 6 years later.

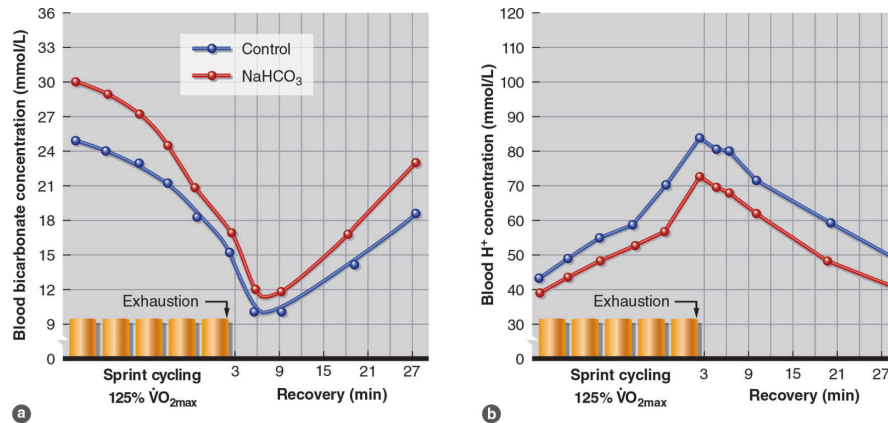


FIGURE 16.2 Concentrations of (a) blood bicarbonate (HCO_3^-) and (b) blood hydrogen ion (H^+) before, during, and after five sprint-cycling bouts with and without ingestion of sodium bicarbonate (NaHCO_3). The fifth sprint was performed to exhaustion. The elevated blood HCO_3^- concentrations caused attenuation of the elevation in blood H^+ , a smaller decline in blood pH, a 42% increase in performance to exhaustion during the fifth sprint, and faster recovery after the sprints.

Adapted by permission from D.L. Costill et al., "Acid-base Balance During Repeated Bouts of Exercise: Influence of HCO_3^- ," *International Journal of Sports Medicine* 5 (1984): 228-231. © Georg Thieme Verlag KG.

Risks Associated with Bicarbonate

Although smaller amounts of sodium bicarbonate have long been used as a remedy for indigestion, many authors studying bicarbonate loading have reported severe gastrointestinal discomfort in some of their subjects, including diarrhea, cramps, and bloating, from the high doses of sodium bicarbonate or citrate required for a buffering effect. These symptoms can be reduced by ingesting ample water and by dividing the total bicarbonate or citrate into five equal parts over a 1 to 2 h period (e.g., five doses of 400 mg for a 70 kg athlete).³⁸

β-Alanine

Consuming fairly large amounts of sodium bicarbonate or sodium citrate increases the buffering capacity of the *extracellular* fluid compartment and is associated with improved high-intensity performance. But is it possible to increase the buffering capacity of the *intracellular* compartment, particularly in muscle cells? Research on β-alanine loading suggests that improved intracellular buffering may be possible.

Proposed Ergogenic Benefits of β-Alanine

β -alanine is an amino acid and as such is used by cells to create a variety of different proteins. One of those proteins is carnosine, a small, simple molecule found in high concentrations in muscle and nerve cells. Carnosine is composed of β -alanine and histidine, another amino acid. Carnosine functions as an intracellular buffer of the hydrogen ions (H^+) produced during intense exercise, especially as a result of lactic acid production.

Demonstrated Effects of β -Alanine

Research shows that β -alanine loading (e.g., 3 g/day for 4 weeks as a loading dose, followed by 1.5 g/day as a maintenance dose) results in increased muscle cell carnosine levels and improved high-intensity performance.³³ In addition, combining β -alanine loading with bicarbonate loading may have an additive effect on performance, perhaps in events lasting an hour or more.⁶ Although some results look promising, other studies report no performance benefit of β -alanine loading, so more research is needed before we have a better understanding of the mechanisms by which, the extent to which, and the consistency with which β -alanine improves performance.

Risks Associated with β -Alanine

More research is required to fully understand the side effects and possible risks associated with β -alanine loading. For example, one study reported that subjects were able to identify the β -alanine treatment because it caused paresthesia (tingling of the skin).¹⁷ Additional research is needed to determine if other side effects are associated with β -alanine loading.

Leucine

Leucine is a *branched-chain amino acid*, as are isoleucine and valine. Leucine is also an essential amino acid that must be supplied by the diet. Soybeans, beef, peanuts, fish, chicken, and almonds are examples of food sources with relatively high leucine content. During recovery from exercise training, adequate food and fluid intake helps muscle cells restore intracellular glycogen levels, reestablish intracellular hydration, repair damaged proteins, and build new proteins.

Proposed Ergogenic Benefits of Leucine

Athletes, coaches, and sport scientists are all interested in ways to optimize these processes, and leucine intake appears to play an important role as a trigger to stimulate muscle protein synthesis as well as acting as a substrate for newly synthesized protein. Leucine is also catabolized by muscle cells, producing metabolites that may also have anabolic properties. For these reasons, leucine is an interesting candidate as a dietary supplement whose ergogenic properties may mimic the muscle-building effects of illegal anabolic agents.

Demonstrated Effects of Leucine

Wilkinson and colleagues⁵⁵ studied the changes in muscle protein synthesis (MPS) and muscle protein breakdown (MPB) in 15 young men at rest in response to the acute ingestion of 3.42 g of either leucine or a metabolite of leucine that has also been proposed as a dietary supplement: β -hydroxy- β -methylbutyrate (HMB). For 2.5 h after ingestion, MPS and MPB were calculated from blood samples and muscle biopsies of the vastus lateralis in combination with leucine and phenylalanine tracers.

The researchers concluded that consuming small doses (<4 g) of leucine and HMB resulted in similar increases in MPS, responses that are comparable to those that have been reported following the consumption of a mixed meal. In addition, HMB ingestion was associated with a decrease in MPB. (MPB was not measured in the leucine treatment group.)

These results lend support to prior reports that leucine and HMB supplementation are associated with increases in MPS. While HMB and leucine both increased the anabolic signaling molecule *mechanistic target of rapamycin* (mTOR), the effect was more pronounced with leucine. The authors concluded that exogenous HMB induces acute muscle anabolism (increased MPS and reduced MPB) but through different mechanisms than the branched-chain amino acid leucine.

Risks Associated with Leucine

As with all dietary supplement research, more studies are needed before a clear picture emerges as to the anabolic potential of leucine and HMB supplementation. However, research does suggest a

benefit, and there appears to be little to no risk associated with supplementation with leucine or HMB or both.

VIDEO 16.1 Presents Nicholas Burd on the role of leucine in muscle repair and growth.



In Review

- An ergogenic aid is any substance or phenomenon that enhances performance. An ergolytic agent is one that has a detrimental effect on performance. Some substances generally thought to be ergogenic are actually ergolytic.
- Although the placebo effect has a psychological origin, the body's physiological and physical responses to a placebo can be quite real. The placebo effect clearly illustrates how effective our mental state can be in altering physiological status.
- Bicarbonate is an important component of the blood's buffering system, needed to maintain normal pH by neutralizing excess acid.
- Bicarbonate (and citrate) loading is proposed to increase the blood's alkalinity, thus increasing the buffering capacity so that more H^+ can be neutralized. This delays the onset of fatigue.
- Bicarbonate ingestion of 300 mg/kg of body weight can delay fatigue and increase performance in high-intensity bouts of exercise lasting approximately 1 to 7 min.
- Bicarbonate (and citrate) loading can cause gastrointestinal distress, including cramping, bloating, and diarrhea.
- β -alanine is an amino acid that is part of the carnosine molecule, a dipeptide that acts as an intracellular buffer (among other functions).
- β -alanine loading is associated with an increase in the carnosine content of muscle cells.

- The impact of β -alanine loading on high-intensity performance is equivocal; some studies report positive effects and others do not.
 - Intake of the essential branched-chain amino acid leucine intake can play an important role as a trigger to stimulate muscle protein synthesis.
-

Caffeine

Caffeine, as with sodium bicarbonate, is not a nutrient, although it is commonly ingested in beverages such as coffees, teas, soft drinks, energy drinks, and energy shots. Caffeine is also a common ingredient in several over-the-counter medications, often in aspirin and other analgesic products. Caffeine is a central nervous system (CNS) stimulant that acts on adenosine receptors in the brain; its sympathomimetic effects are similar to those of **amphetamines**, although considerably weaker. As a stimulant capable of improving exercise performance, caffeine deserves special attention, in part because it is one of the most widely consumed drugs in the world.

Proposed Benefits of Caffeine

As with sympathomimetic amines, caffeine ingestion is linked to improved alertness, concentration, reaction time, and perceived energy level. Caffeine is known to have metabolic effects on adipose tissue and skeletal muscle as well as on the CNS.

Demonstrated Effects of Caffeine

Because of its effects on the CNS, the general effects of caffeine include

- increased mental alertness,
- increased concentration,
- elevated mood,
- decreased fatigue and delayed onset,
- decreased reaction time (i.e., faster response),
- enhanced catecholamine release,
- increased free fatty acid mobilization, and
- increased use of muscle triglycerides.

The ergogenic benefits of caffeine were first reported by scientists in the late 1970s.^{19,35} Caffeine ingested before exercise increased

endurance times in fixed-pace work bouts and decreased times in fixed-distance tasks.

Although several subsequent studies were unable to replicate these results, recent studies have consistently demonstrated the ergogenic effects of caffeine ingestion.^{30,39} It was initially thought that this improvement was the result of an increased mobilization of free fatty acids, sparing muscle glycogen for later use. But the mechanisms by which caffeine improves endurance performance appear to be more complex, since glycogen sparing does not always occur, and also because caffeine improves performance in tasks in which muscle glycogen is not limiting. A growing number of studies now demonstrate that caffeine has its effect directly on the CNS.⁴⁹ It is now well documented that caffeine lowers the perception of effort, allowing the athlete to perform at a higher intensity with the same perceived effort. The dose of caffeine that is commonly associated with improved performance is 3 g/kg body weight consumed before exercise. That equates to about 200 mg of caffeine for a 70 kg athlete. As a single dose, that is greater than the amount in a 12 oz energy drink (150 mg) or a can of cola (40 mg), but similar to the amount in a large serving of strong coffee.

RESEARCH PERSPECTIVE 16.3

Caffeine Use in Cycling

Caffeine can enhance endurance exercise performance and short-term high-intensity anaerobic exercise performance, as discussed in this chapter. However, the mechanism underlying caffeine's ergogenic effects are not completely understood. In recent years, focus has shifted from caffeine's promotion of fat oxidation (glycogen sparing) to its actions within the central and peripheral nervous systems, which could alter sensations of effort and muscle pain. Yet few studies have systematically assessed neuromuscular function (maximal voluntary contraction, motor-unit recruitment, and muscle contractile properties), muscle pain, perceived effort, and endurance performance after caffeine consumption.

Investigators recently tested two hypotheses: (1) caffeine would improve cycling performance because of increased strength due to enhanced motor-unit recruitment and (2) caffeine would have no effect on the perception of pain and effort during leg cycling performance.¹¹ Using a novel study design that incorporated both leg cycling and arm crank cycling allowed investigators to manipulate caffeine's effects on force production and motor-unit

recruitment. Caffeine ingestion enhanced strength and motor-unit recruitment in the knee extensors but not the elbow flexors. Caffeine ingestion also reduced muscle pain and ratings of perceived exertion during moderate-intensity exercise, but this effect disappeared at high exercise intensities, like those exerted during all-out time trials. Lastly, caffeine ingestion improved endurance performance during leg cycling but not arm crank cycling. These findings cast doubt on the ability of caffeine to be hypoalgesic (decrease the sensitivity to pain) during high-intensity exercise. Instead, the data support the concept that enhanced strength is a likely mechanism, at least in part, by which caffeine is ergogenic.

Caffeine has also been shown to improve performance in sprint and strength types of activities and in high-intensity team sports. Unfortunately, fewer studies have investigated the mechanisms by which caffeine might improve performance in high-intensity activity. However, in addition to CNS stimulation, caffeine might facilitate calcium exchange at the sarcoplasmic reticulum and increase the activity of the sodium–potassium pump, better maintaining the muscle membrane potential.³⁰

Risks Associated with Caffeine Use

In people who are not accustomed to using caffeine, who are sensitive to it, or who consume high doses, caffeine can produce nervousness, restlessness, insomnia, headache, gastrointestinal problems, and tremors. Caffeine also acts as a mild diuretic at rest, but this effect plays no significant role during exercise because renal blood flow and urine production decreases with exercise. Regular caffeine use does not jeopardize daily hydration, especially among habitual caffeine users, because the diuretic effect of caffeine is muted and people generally consume enough water and other beverages to maintain 24 h hydration status. Caffeine can disrupt normal sleep patterns in some people, contributing to fatigue. Caffeine is also physically addictive; abrupt discontinuation of intake can result in headache, fatigue, irritability, and gastrointestinal distress. At one time, caffeine was on the WADA list of prohibited substances, but in 2004 it was removed from the list, although its presence in urine samples is still monitored.

Cherry Juice Polyphenols

Fruit and vegetable juices contain hundreds, if not thousands, of compounds with biological activities and health-related effects that are poorly understood simply because of a lack of research. Of interest to athletes and coaches is emerging research on how tart cherry juice might be a potential nutrition intervention with analgesic (pain-reducing) effects that could speed recovery from training bouts causing muscle damage and related pain.

Proposed Benefits of Cherry Juice Polyphenols

Cherries, much like blueberries, cranberries, raspberries, purple grapes, pomegranate, acai berries, and other dark fruits, contain polyphenolic compounds such as flavonoids, secondary plant metabolites that provide the pigmentation characteristics of the fruit and also help protect the fruit from microbes and insects. Daily polyphenol intake is one of the reasons dietitians recommend several servings of fruits and vegetables each day. (There are estimated to be more than 4,000 different flavonoids in foods, and we typically eat about 1,500 mg of flavonoids every day.) Eating dark fruits such as cherries exposes our cells to flavonoids such as gallic acid, kaempferol, quercetin, resveratrol, and other compounds with antioxidant and anti-inflammatory properties. Unlike taking an antioxidant supplement, eating a dark fruit provides a large array of micronutrients that play important roles in the life cycle of plants. Whether or not that array of micronutrients (or single micronutrients) provides meaningful benefits beyond good nutrition is a question that scientists are beginning to address. The premise for the use of cherry juice as an ergogenic aid is that the anti-inflammatory and antioxidant characteristics of cherries may have a beneficial effect for pain reduction following muscle-damaging exercise.

Demonstrated Effects of Cherry Juice Polyphenols

Studies in which subjects consumed cherry juice or placebo for a week or more before a bout of eccentric exercise have reported a reduction in oxidative stress, strength loss, and subjective perceptions of muscle pain associated with the cherry juice treatment.^{12,18,37} These studies suggest that cherry consumption might be associated with real benefits, associations that will eventually be supported or rejected through further research. At this time, it is

impossible to state with any confidence whether cherry juice consumption is associated with consistent analgesic effects.

Possible Risks Associated with Cherry Juice Polyphenols

In terms of basic nutrition, cherries are wholesome, healthy, and packed with a wide variety of good-for-us nutrients. For that reason, there is no obvious downside to including cherries (or cherry juice) in an athlete's diet; and, optimistically speaking, the anti-inflammatory upside of cherries just might do muscles some good. However, it will take years of research and dozens of studies to determine whether or not that optimism can be scientifically justified.

Creatine

Creatine has become a popular ergogenic aid among athletes, especially those participating in team sports and other activities that involve repeated, high-intensity movements. Creatine is a molecule found inside every muscle cell. The body is capable of synthesizing creatine in the liver, with additional creatine supplied by the diet (mostly from meat and fish). Consuming creatine supplements to boost daily creatine intake has been shown to increase the creatine content of muscles.

Proposed Benefits of Creatine

The primary use of creatine is based on its role in skeletal muscle, where approximately 60% of the total creatine content is in the form of phosphocreatine (PCr). Increasing the creatine content of skeletal muscle cells through creatine supplementation is theorized to increase muscle PCr levels, thus enhancing the adenosine triphosphate (ATP)-PCr energy system by better maintaining muscle ATP levels. This would theoretically enhance peak power production during intense exercise and possibly facilitate recovery from high-intensity exercise. Creatine also serves as an intracellular buffer, helping to regulate acid–base balance, and plays a role in oxidative metabolic pathways. These are two other possible ways in which creatine loading might improve performance.

Demonstrated Effects of Creatine

Because of the popularity of creatine supplementation in the 1990s and the wide-ranging claims of its ergogenic properties, the American College of Sports Medicine (ACSM) published a consensus statement, “The Physiological and Health Effects of Oral Creatine Supplementation,” in 2000.² A group of scientists reviewed the research literature on creatine and performance to develop a list of conclusions supported by scientific research. The following conclusions were drawn:

- Creatine supplementation can increase muscle PCr content, although not in all individuals.
- Exercise performance in short periods of intense, high-power exercise can be enhanced, particularly with repeated bouts, consistent with the role of PCr in this type of activity.
- Maximal isometric strength, the rate of maximal force production, and maximal aerobic capacity are not enhanced by creatine supplementation.
- Creatine supplementation often leads to weight gain within the first few days, likely attributable to water accumulation with creatine uptake in the muscle.
- In combination with resistance training, creatine supplementation is associated with greater gains in strength, perhaps in relation to the increased ability to train at higher intensities.
- The high expectations for performance enhancement exceed the true ergogenic benefits of creatine loading.

RESEARCH PERSPECTIVE 16.4

Creatine Supplementation Plus Resistance Exercise to Prevent Sarcopenia

Sarcopenia is the loss of muscle mass (myopenia) and strength (dynapenia) that occurs with normal aging. It is characterized by type II muscle fiber atrophy, loss of myofibers, and increased intramuscular lipid and connective tissue. Resistance training increases both muscle mass and muscle strength and is therefore an important countermeasure to sarcopenia. In addition to resistance training, protein consumption also stimulates muscle protein synthesis and may be an additional strategy for mitigating age-related losses in muscle mass and strength. Does the use of protein supplementation in the

form of creatine and resistance training together have a greater effect on muscle strength and functional performance than either resistance training or creatine supplementation alone? In 2014, investigators performed a meta-analysis (systemically combining findings from several different studies) to determine whether the addition of creatine supplementation to resistance training improves body composition and increases strength and functional performance in older adults.²¹

Ten studies with a total of 357 participants met the inclusion criteria and were included in the meta-analysis. As hypothesized, in older adults, creatine supplementation during resistance training (≥ 6 weeks) improved body composition, strength, and functional performance to a greater extent than resistance training alone. The mechanisms by which creatine improves muscle strength are likely multifactorial and may potentially include increased phosphocreatine energy stores, increased phosphocreatine synthesis, and reduced muscle damage—all of which enhance the ability to perform resistance training, thereby having a greater effect on body composition, strength, and functional performance.

When considered collectively, the results of this meta-analysis support a role for concurrent resistance training and creatine supplementation in older adults to slow sarcopenia.

Subsequent scientific reviews have been published since the release of the ACSM consensus statement and are generally in agreement. Creatine is one of the few supplements that, in combination with resistance exercise, is associated with increases in both fat-free mass and strength.⁴⁰ With respect to improving athletic performance, the studies are mixed. This is likely due to two factors: the physiological demands of the sport or event and the individual variability in response to the supplement. Performance enhancement is more likely to occur in sports involving brief periods of repeated high-intensity exercise.⁷ Concerning individual variability, we discussed in [chapter 9](#) the principle of individuality—the fact that there are high responders and low responders to any given intervention. In studies involving only a few subjects (e.g., <10), it is possible that more high than low responders are represented in the study sample or vice versa.

Risks Associated with Creatine

There do not appear to be any short-term health risks associated with appropriate levels of creatine supplementation, a conclusion based on a number of research articles and the widespread use of creatine

by athletes. One remaining note of caution relates to long-term creatine supplementation, especially in young, growing athletes, because there have been no long-term studies on the topic. It may well be that creatine supplementation presents no long-term health risks, but verification of that conclusion awaits further research. There are several reported cases of kidney damage in young athletes who used massive doses of creatine.

Nitrate

Inorganic nitrate (NO_3^-) is a molecule found in all plants (as part of the nitrogen cycle that you might remember from high school biology class); vegetables such as spinach, celery, and beets contain higher levels of nitrate than other vegetables. Bacteria on the tongue (and other enzymes in the body) convert nitrate to nitrite (NO_2^-). Cells can then convert nitrite to nitric oxide (NO). Nitric oxide can also be produced from the amino acid arginine via nitric oxide synthase enzyme (NOS). Citrulline, another amino acid and a precursor to arginine, is included in some dietary supplements as an endogenous source of NO. Our diets provide an estimated 50% of the NO produced in the body, and dietary intake of nitrate represents one way in which NO production might be increased. Nitric oxide molecules play a critical role in a wide variety of functions, including control of blood flow, white blood cell chemotaxis, mitochondrial function, and cellular apoptosis.

Proposed Benefits of Nitrate Ingestion

Because NO affects muscle blood flow and mitochondrial function, scientists have been interested in determining if ingestion of nitrate, arginine, and citrulline (and a few other compounds) is associated with ergogenic benefits. For example, if nitrate ingestion results in increased delivery of oxygen and nutrients to active skeletal muscles, that response could be the basis for improved performance in a variety of exercise settings ranging in intensity and duration.

Nitrate (and nitrite) ingestion is also associated with a reduction in resting systolic blood pressure and other positive cardiovascular changes that might alter the course of cardiovascular aging and therefore have important implications for overall cardiovascular health. As with nitrate ingestion and performance, this is a promising

area of research that requires more studies using subjects of both sexes and a range of ages as well as a variety of intake doses consumed over varying durations and intensities of exercise.

Demonstrated Effects of Nitrate Ingestion

Early research on nitrate ingestion (often in the form of beet juice) showed performance benefits, a finding that many other studies—although not all—have corroborated. As is always the case with research on ergogenic aids, many studies are needed before a clear picture emerges, and nitrate ingestion is no exception. However, a growing body of well-done research supports an ergogenic effect with nitrate ingestion, at least in untrained or modestly fit young male subjects. It appears that the effect of nitrate ingestion might be considerably less than the benefits associated with being highly fit, because studies using very fit subjects often report no benefit of nitrate ingestion.⁹

Research has consistently shown that when subjects consume beet juice for a week before an exercise challenge, or even a few hours before exercise, plasma nitrite concentration rises, resting blood pressure decreases, exercise oxygen consumption falls slightly, and exercise performance is significantly improved. It is thought that these responses are related to enhanced muscle cell contractility, improved efficiency of ATP production in mitochondria, and increased blood flow to muscle cells. Most of these studies have examined a single dose of beet juice; little work has been done to determine the dose of beet juice (i.e., the dose of nitrate) that is associated with the greatest responses.

In one study, scientists reported that eight male subjects who ingested 500 ml (16 oz) of beet juice per day for 6 days (an average of 486 mg of sodium nitrate per day) improved their time to exhaustion on a cycle ergometer test even though oxygen consumption was reduced, a finding consistent with other studies.⁵ The hypothesis is that nitrate ingestion increases NO production and improves mitochondrial efficiency for ATP production, reducing oxygen consumption even as performance improves. It appears that acute consumption of beet juice before exercise might also be ergogenic.⁵²

In another study,⁵⁸ 10 young, recreationally active men ingested 70, 140, or 280 ml (roughly 2.5, 5.0, and 10.0 oz) of beet juice at rest and immediately before exercise. Following beet juice ingestion, plasma nitrate concentration rose in a dose-dependent manner, with peak levels at 2 to 3 h postingestion. The lowest volume of beet juice did not affect exercise responses, but the 140 and 280 ml volumes were associated with lower average steady-state oxygen consumption (by 1.7% and 3.0%, respectively) and improved exercise performance (by 14% and 12%, respectively) compared to the control treatment (water). Performance was assessed as time to failure on a cycle ergometer as the subjects repeated bouts of moderate- and high-intensity exercise until they could not maintain a preset pedaling cadence.

Although the exact mechanisms by which nitrate ingestion alters exercise performance are not yet entirely clear, many research studies have reported similar results, at least with recreationally active subjects. More work is needed to corroborate these early findings, further understand dose–response relationships, identify mechanisms of action, and provide additional recommendations to coaches and athletes interested in using beet juice or other nitrate sources as part of their training and competitive strategies.

Possible Risks Associated with Nitrate Ingestion

Adverse health consequences do not appear to be associated with nitrate intake, with the possible exception of people with heart disease who are taking prescription medications (e.g., nitroglycerine) that also affect NO metabolism. However, little is known about the possible side effects of consuming large amounts of nitrate supplements.

In Review

- Caffeine can enhance performance in endurance sports and may even be of benefit in activities of much shorter duration (e.g., 1 to 6 min). However, some athletes may experience a negative reaction to caffeine, in which case caffeine could be ergolytic.
- There is some evidence that consuming tart cherry juice is associated with reduced pain and faster recovery of strength from eccentric exercise compared to a placebo treatment.

- Creatine supplementation appears to have ergogenic benefits, particularly for improving performance in repeated, high-intensity exercise bouts between 30 and 150 s in duration.
 - Creatine supplementation has been shown to increase muscle creatine levels and increase performance in sports involving brief, repeated periods of high-intensity exercise.
 - Ingesting nitrate (usually in the form of beet juice) is associated with improved exercise performance in sedentary or recreationally active subjects but is less commonly evident in well-trained subjects. Nitrate ingestion has been shown to lower systolic blood pressure and may have other benefits to cardiovascular health.
-

Anti-Doping Codes and Drug Testing

Performance-enhancing drugs (PEDs), used by athletes who choose to cheat in order to gain an unfair advantage over their competitors, have a long history in sport. Several of these prohibited substances and techniques are discussed in more detail later in the chapter. Increased PED use among athletes has led to the formation of regulatory organizations, increased testing of supplement products and ingredients for contamination, and the proliferation of designer drugs. In this section, we introduce prohibited substances from this regulatory perspective.

The World Anti-Doping Code

In 1968, the International Olympic Committee (IOC) began drug testing athletes during the Summer and Winter Games. As concerns over the use of PEDs grew through the 1970s and 1980s, an IOC-led initiative established the World Anti-Doping Agency (WADA; www.wada-ama.org) in 1999. WADA is composed of and funded by the Olympic Movement and various governments.

The actions of WADA are guided by the World Anti-Doping Code, adopted in 2003. The Code is the core document that provides the framework for harmonized anti-doping policies, rules, and regulations within sport organizations and among public authorities. More than 600 athletic governing organizations have now adopted the Code.

An important component of the Code is the Prohibited Substances List, which is updated annually. Substances or practices are

considered for inclusion if they meet two of the following three criteria:

1. There is scientific or medical evidence that the substance or practice has the potential to enhance, or has been shown to enhance, sport performance.
2. There is scientific or medical evidence that use of the substance or practice presents a health risk to athletes.
3. The World Anti-Doping Agency determines that the substance or practice violates the spirit of sport.

Possession, trafficking, or administration of PEDs and banned practices are also considered violations of the Code. Banned practices include tampering with or substituting urine or blood samples. For instance, athletes have tried to substitute someone else's urine as their own or have attempted to add proteases or other substances to a urine sample to mask banned substances. Intravenous infusions of saline, glucose, and other solutions are also prohibited in competitive settings. Even gene doping is now a banned practice in anticipation of what future technologies might allow.

Compliance with the Code is monitored through a testing program administered in conjunction with various sport governing bodies. Athletes must report their whereabouts and be available for randomized testing. Those who test positive for a banned substance or practice may be subject to a variety of sanctions ranging from increased monitoring up to a lifetime ban from their sport.

The Code is based on the principle of strict liability; that is, athletes are responsible for any substance in their body, even if they were unaware of it. In other words, even if an athlete accidentally ingests a banned substance, the athlete is held fully responsible. Athletes who have medical conditions that require the medicinal use of a banned substance (for example, an athlete who is prescribed an asthma medication) may apply for a **therapeutic use exemption**. Athletes who believe that a positive test or sanction is incorrect or unfair may appeal the result through their sport's governing body and the Court of Arbitration for Sport.

Other sport organizations such as the NCAA, the National Football League, and NASCAR have their own banned substance lists but all are similar in many respects to the WADA list. The IOC, the United

States Olympic Committee (USOC), the International Association of Athletics Federations (IAAF), and the NCAA all publish extensive lists of banned substances, most of which are pharmacological agents. The IOC and the USOC use the standards established by WADA. In the United States, these standards are administered through the United States Anti-Doping Agency (USADA; www.usada.org). The list of banned substances is updated annually. [Table 16.1](#) lists prohibited substances as of 2017 and the mechanisms of action by which these ergogenic aids have been proposed to work.

Each athlete, coach, athletic trainer, and team physician must know which drugs are prescribed for and taken by the athlete, and they must check these drugs periodically against the relevant listing of banned substances because lists change frequently. Athletes who require prescription medications can apply for and be granted therapeutic use exemptions so they can continue to take their medications without risk of running afoul of anti-doping regulations.

Contamination of Dietary Supplements

Many athletes take one or more types of nutritional supplements, ranging from vitamin and mineral tablets to elaborate mixtures of plant parts and herbs. Most athletes assume that they are ingesting a substance that exactly matches the ingredients listed on the product container. Unfortunately, that may not be the case, since accidental or purposeful contamination of dietary supplements with banned substances has occurred repeatedly. Although most supplement manufacturers are diligent in ensuring the purity and potency of their products, some are not. This inconsistency requires all supplement consumers to follow a *caveat emptor*—buyer beware—approach to purchasing and consuming supplements.

Starting in 1999, researchers have investigated the potency and purity of selected sport supplements. The findings were sobering. In some cases, product potency varied from negligible values to up to 150% of the dose listed on the product label. In addition, many supplements were confirmed to be contaminated with prohibited substances²⁷ that could lead to positive doping results and subsequent severe penalties for athletes. Common contaminants have included anabolic steroids, stimulants, and diuretics. Numerous studies have substantiated the extent and the critical nature of this

problem. The bottom line: Athletes are responsible for what they ingest; those who use supplements are taking what could be an extremely high risk.

Supplement consumers—and especially athletes—should be concerned about three critical aspects with any supplement they are considering.

1. Purity—does the supplement contain any harmful or prohibited substances?
2. Potency—does the supplement actually contain the amounts of ingredients listed on the label?
3. Efficacy—does the supplement actually provide any benefits?

Unfortunately, there are no easy answers to any of these questions. Some supplement manufacturers enlist third parties to analyze their products for potency and purity and display a certification mark on the product label. Doing so reflects a good-faith effort on the part of the manufacturer to assure consumers that their manufacturing processes and product formulations adhere to the most current standards. But as rigorous as testing for banned substances can be, it is impossible to ensure that any product is completely free of banned substances because it is impossible to test for all banned substances. However, reputable manufacturers that make the extra effort and incur the expense of third-party testing are doing the right thing on behalf of their consumers.

TABLE 16.1 WADA 2017 Prohibited Substances List, with Possible Mechanisms of Action (Where Appropriate)

Agent	Influence heart, blood, circulation, and aerobic endurance	Increase oxygen delivery	Supply fuel for muscle and general muscle function	Act on muscle mass and strength	Result in weight loss or weight gain	Counteract or delay onset or sensation of fatigue	Counteract central nervous system inhibition	Aid in relaxation and stress reduction	Speed recovery
Anabolic agents				✓	✓				✓
Peptide hormones	✓	✓		✓	✓				✓
β-2 agonists	✓					✓	✓		
Hormone and metabolic modulators	✓		✓		✓	✓	✓		
Diuretics and other masking agents	✓			✓	✓				
Blood manipulation	✓	✓	✓			✓			
Chemical and physical manipulation (tampering with samples)									
Chemical and physical manipulation: large intravenous infusions	✓	✓	✓			✓			✓
Gene doping	✓	✓	✓	✓	✓	✓	✓	✓	✓
Stimulants	✓		✓			✓	✓		✓
Narcotics						✓		✓	
Cannabinoids								✓	
Glucocorticosteroids	✓		✓	✓		✓			
Alcohol								✓	
β-blockers	✓					✓		✓	

Designer Steroids and Drug Testing

In an effort to circumvent sport drug testing, some chemists have become adept at synthesizing variations of steroids—or their metabolic precursors—that cannot be detected by current testing procedures. That does not mean that these illicit compounds are undetectable—all compounds can be detected by the right instruments and analytical techniques. But sport drug testing typically analyzes for an established array of steroids, and if a new steroid is not in that array, it will not be detected until someone begins to look for it. This is exactly what happened with tetrahydrogestrinone (THG)—the powerful designer steroid illegally used by athletes in the 1990s. Tetrahydrogestrinone remained undetected by sport drug

testing until a sample of the substance was analyzed by forensic toxicologists and subsequently added to the WADA banned substance list.

The World Anti-Doping Agency recognizes two classes of steroids: (1) endogenous steroids such as **testosterone**, estrogen, and their metabolic precursors including epitestosterone, androstenedione (Andro), and dehydroepiandrosterone (DHEA); and (2) exogenous steroids—in other words, synthetic steroids such as nandrolone, trenbolone, methyltestosterone, THG, and other designer steroids.

Most sport drug testing is accomplished with urine samples; in the case of steroids, the test methodology requires steps to extract, separate, and detect steroid metabolites in the urine. After the urine sample has been treated to make the metabolites measurable, the sample is typically run through gas or liquid chromatographs to separate the compounds and through mass spectrometers to identify the compounds.

In addition to random drug testing, the Athlete Biological Passport (ABP) will see more use in future years. The ABP is a way to determine a young athlete's normal profile for measures such as testosterone, epitestosterone, the ratio between those two steroids (T/E ratio), hemoglobin, hematocrit, and other parameters that can be used to establish a normal baseline for individual athletes. Major changes in those parameters reflect possible use of banned substances. For example, the T/E ratio has long been used as a marker of steroid use. The average T/E ratio in men is 1:1. Use of illegal steroids increases that ratio. Because there is a large variation in T/E ratio among people, sport governing bodies have adopted cut-off values such as 4:1 or 6:1 to allow for that normal variation. Of course, this does not help athletes whose normal T/E ratio is 10:1, nor does it prevent athletes from using a combination of testosterone and epitestosterone to benefit from an anabolic effect without altering the T/E ratio. However, it is now possible to distinguish exogenous testosterone from endogenous testosterone, another analytical weapon that can be used to catch cheaters.

In Review

- The World Anti-Doping Code, first adopted in 2003, is the core document that provides the framework for harmonized anti-doping policies, rules, and regulations within sport organizations and among public authorities. More than 600 athletic governing organizations have now adopted the Code.
 - Substances are considered for inclusion on the WADA banned substance list if they meet any two of the following three criteria based on scientific evidence: (1) They have the potential to enhance, or have been shown to enhance, sport performance, (2) they present a health risk to athletes, or (3) they violate the spirit of sport.
 - A substantial risk is associated with using nutritional supplements due to potential contamination of the products and the challenges associated with ensuring that supplements are free of banned substances.
-

Prohibited Substances and Techniques

Table 16.1 introduced a list of the substances, agents, and methods included in the WADA 2017 Prohibited Substances List. In this section, we consider the following prohibited substances and practices in detail:

- Stimulants
- Anabolic steroids
- Human growth hormone
- Diuretics and other masking agents
- β -blockers
- Blood doping

Stimulants

Stimulants include amphetamines and related compounds. Stimulants are often referred to as *sympathomimetic* amines, which means that their activity mimics that of the sympathetic nervous system. For many years, stimulants have been used as appetite suppressants in medically supervised weight loss programs. During World War II, army troops used amphetamines to combat fatigue and to improve endurance. Stimulants are now used to treat attention deficit hyperactivity disorder (ADHD) to activate parts of the brain that help reduce the symptoms associated with ADHD and similar disorders. Long ago, amphetamines found their way into sport. Other

sympathomimetic amines such as **ephedrine** and **pseudoephedrine** have been used as ergogenic aids. Ephedrine is derived from herbs (often known as ma huang) and has been used as a decongestant and as a bronchodilator in the treatment of asthma. Pseudoephedrine is used in over-the-counter medications primarily as a decongestant and in the illicit manufacturing of methamphetamine.

Proposed Benefits of Stimulants

Stimulants have physiological and psychological effects that reduce appetite, increase concentration and alertness, boost metabolic rate, and reduce the sensation of fatigue. Athletes who have used amphetamines report a sense of indestructibility, which they feel spurs them on to higher performance levels. Others rely on the sympathetic stimulation to raise metabolic rate and promote fat loss.

In terms of actual performance, amphetamines are thought to help athletes run faster, throw farther, jump higher, and delay the onset of fatigue. Similar claims and expectations have been associated with the use of ephedrine and pseudoephedrine.

Demonstrated Effects of Stimulants

Not surprisingly, some studies report that amphetamines have no effect on performance, others demonstrate a strong ergogenic effect, and still others conclude that amphetamines have an ergolytic effect. As potent CNS stimulants, amphetamines do increase the state of arousal, which leads to a sense of increased energy, self-confidence, and faster decision-making. Athletes who take amphetamines experience a decreased sense of fatigue, along with increased heart rate, systolic and diastolic blood pressure, and blood flow to skeletal muscles, as well as an elevation of blood glucose and free fatty acids.

The key question is, do these effects aid physical performance? Although studies are not in total agreement, research that has used better experimental designs and controls shows that amphetamine ingestion is associated with the types of responses that would be expected from a CNS stimulant, including

- weight loss;
- improved reaction time, acceleration, and speed;
- greater strength, power, and muscular endurance;

- possibly improved aerobic endurance, although not $\dot{V}O_{2\max}$;
- higher maximum heart rate and peak lactate concentrations at exhaustion;
- improved focus; and
- enhanced fine motor coordination.

The results are not as clear for less-potent stimulants such as synephrine, ephedrine, and pseudoephedrine. While studies have shown small improvements in markers of athletic performance with use of these substances, the current thinking is that performance benefits are inconsistent for speed, strength, power, and endurance.^{1,16}

Risks Associated with Stimulants

Deaths have been attributed to excessive amphetamine and ephedrine use, likely because heart rate and blood pressure are increased, placing greater stress on the cardiovascular system. Stimulants can trigger cardiac arrhythmias in susceptible individuals. Also, rather than delaying the onset of fatigue, amphetamines likely delay the sensation of fatigue, enabling some athletes to push dangerously beyond the normal limits of exhaustion.

Amphetamines can be psychologically addictive because of the euphoria and energized feelings they cause. But the drugs also can be physically addictive if taken regularly, and a person's tolerance to them builds with continued use, requiring increasingly larger doses over time to obtain the same effects. Amphetamines also can be toxic. Extreme nervousness, acute anxiety, aggressive behavior, and insomnia are side effects of regular use. Ephedrine has side effects similar to those of amphetamines and is associated with risks of cardiovascular events and heat illness.

Anabolic Steroids

Anabolic steroids are prohibited substances because of their well-established ergogenic benefit of increasing muscle mass beyond what can be normally achieved by the combination of strength training and diet. Steroids and similar substances have been used by athletes in sports ranging from endurance cycling to baseball to bodybuilding. Many male sex hormones have both muscle-building

and masculinizing effects and are therefore referred to as anabolic-androgenic steroids. The use of steroids and other hormonal agents as ergogenic aids began in the late 1940s or early 1950s, if not sooner. Anabolic agents are banned for all sports, and the medical risks associated with their use are high.

Anecdotal stories suggest that anywhere from 20% to 90% of athletes in certain sports are using, or have used, anabolic steroids. Scientific surveys, however, suggest a much lower estimate of 6% across sports.⁸ Anabolic steroid use has even been reported in the general high school population in the United States, varying from 4% to 11% in boys and up to 3% in girls.^{16,43}

Steroids are a large class of substances that include male and female sex hormones such as testosterone and estradiol. Interestingly, cholesterol is also classified as a steroid. Although cholesterol has no anabolic or androgenic effects, its structure is similar to that of all steroid hormones. Plants also produce steroids (phytosterols) that, like cholesterol, do not interact with androgen receptors in cells and therefore have no anabolic effects. Steroids that do interact with androgen receptors, however, accelerate growth by increasing the rate of bone maturation and the development of muscle mass. For years, anabolic steroids have been given to children with delayed growth patterns to normalize their growth curves. Dozens of steroids have been synthesized with slightly altered chemical structures that reduce the androgenic and anabolic properties of the hormones. As a result, synthetic steroid hormones differ in their anabolic and androgenic (masculinizing) effects.

Proposed Benefits of Anabolic Steroids

Steroid administration is known to increase muscle mass, and therefore strength, and to reduce fat mass. For those reasons, steroids are understandably classified as a PED. In addition, anabolic steroids have been postulated to facilitate recovery from exhaustive training bouts, allowing athletes to train hard on subsequent days.

Athletes who take steroids have become very good at beating the system. Diuretics and other masking agents and techniques have been used to reduce the risk of detection. And chemists continue to develop designer steroids to help cheating athletes avoid detection. Randomized drug testing (both in competition and out of season),

coupled with improved analytical techniques, has increased the likelihood that PED users will be caught. Those actions, along with improved education efforts, will hopefully be successful in deterring most athletes from using steroids.

Demonstrated Effects of Anabolic Steroids

The ergogenic benefits of anabolic steroids are clear. A strong dose–response relationship between steroids and lean body mass, muscle mass, and strength has now been demonstrated, making steroids one of the most powerful prohibited ergogenic aids. However, that was not always the case. Results of early studies on anabolic steroids were almost evenly divided among those studies that showed no significant change in body size or physical performance and those that reported significant improvements in muscle mass and strength.

One basic problem with almost all research conducted in this area is the researcher's inability to use the large drug dosages commonly consumed by athletes. It is estimated that some athletes use 5 to 20 times (or more) the recommended maximum daily medicinal dosage.³¹ However, some researchers have been able to observe athletes when they are taking high doses of steroids and when they are off them, aiding our understanding of how anabolic steroids affect muscle size and strength.

In one of the first studies involving athletes who were illegally taking steroids, the effects of relatively high doses were observed in seven male weightlifters.³² Two treatment periods, each lasting 6 weeks, were separated by a 6-week interval without treatment. Half the subjects received a placebo during the first treatment period and then the steroid during the second treatment period. The other half received the medications in reverse order: steroid first, then placebo. When the data from all subjects were analyzed, results showed that while on the steroid, the weightlifters had significant increases in

- body mass and fat-free mass,
- total body potassium and total body nitrogen (markers of fat-free mass),
- muscle size, and
- leg strength.

These increases did not occur during the placebo period. Results of this study are summarized in [figure 16.3](#).

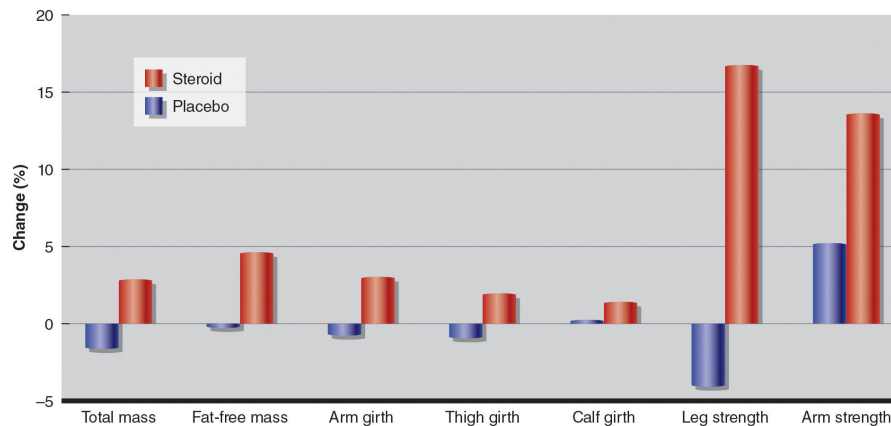


FIGURE 16.3 Percentage changes in body size, body composition, and strength when athletes used anabolic steroids or a placebo.

Adapted from Hervey et al. (1981).

In a second study, Forbes²⁶ observed body composition changes in a professional bodybuilder and a competitive weightlifter. Both were on self-prescribed high doses of steroids. The bodybuilder continued on the high dose for 140 days and the weightlifter for 125 days. Fat-free body mass increased an average of 19.2 kg (42.3 lb), and fat mass decreased almost 10 kg (22 lb). Forbes²⁶ plotted the results of a number of studies that used different dosages ([figure 16.4](#)). He observed that minimal increases of 1 to 2 kg (2.2-4.4 lb) in fat-free body mass occur with low doses of anabolic steroids. But with high doses, fat-free body mass increases markedly. These results suggest a threshold level for steroid doses, with only regular high doses resulting in substantial increases in fat-free body mass. Similarly, brief increases in testosterone, such as those that might occur after strength training or protein intake, do not appear to have a major effect on muscle mass or strength.

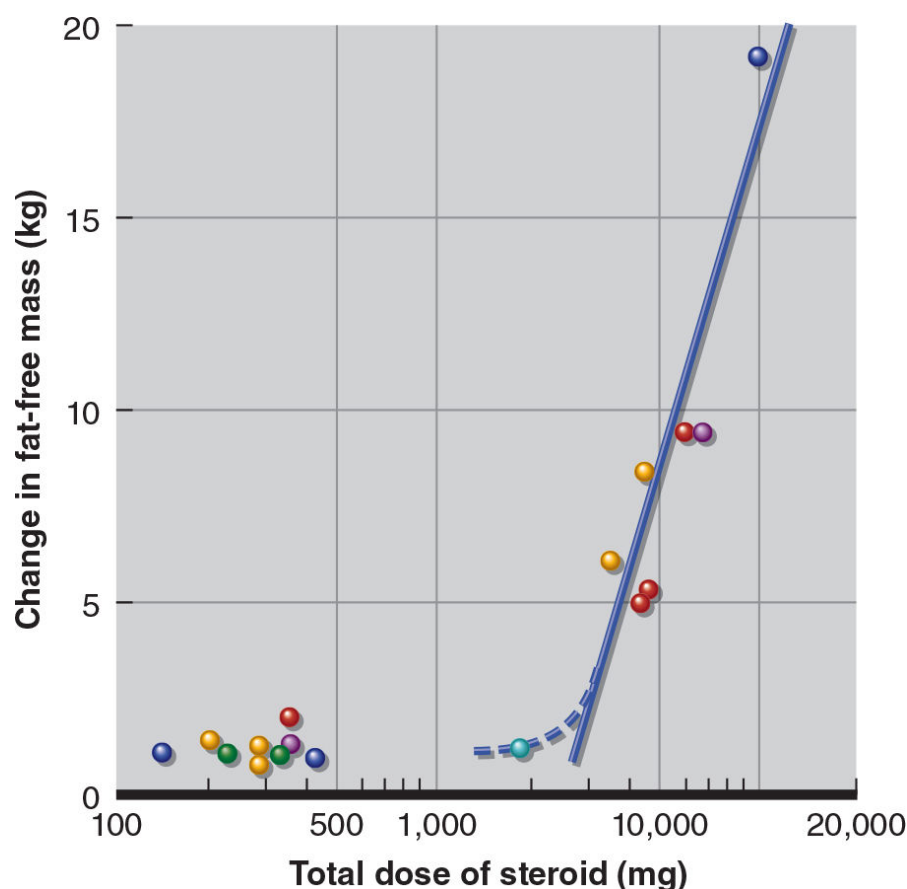


FIGURE 16.4 The relationship between the total dose of steroid (mg/day) and the change in fat-free mass in kilograms. The colors represent different anabolic steroid drugs. Steroid dose plotted logarithmically.

Adapted by permission from G.B. Forbes, "The Effect of Anabolic Steroids on Lean Body Mass: The Dose Response Curve," *Metabolism*, 34 (1985): 571-573, Copyright 1985, with permission from Elsevier. <http://www.sciencedirect.com/science/journal>

A third study looked at supraphysiological doses of testosterone in relation to muscle size and strength in men who were nonathletes but were experienced with weightlifting.¹⁰ Forty-three men completed the study and were assigned to one of the following groups: placebo with no exercise, placebo with exercise, testosterone with no exercise, and testosterone with exercise. The men received either 600 mg of testosterone enanthate or placebo intramuscularly each week for 10 weeks. The exercise groups lifted weights 3 days each week for 10 weeks. Body composition was measured by underwater weighing, triceps and quadriceps muscle size by magnetic resonance imaging, and arm and leg strength by 1RM bicep curl and squat testing. In the no-exercise groups, testosterone injections resulted in increased arm

and leg size and strength. Subjects who strength trained and received testosterone injections had the greatest increases in arm and leg size, mass, and strength (see [table 16.2](#)).

The increase in muscle mass is generally associated with increases in the cross-sectional areas of both type I and type II muscle fibers and an increase in the number of muscle nuclei. These increases are dose dependent and likely due to increased muscle protein synthesis resulting from continued stimulation of muscle androgen receptors by the exogenous steroids.²⁵

The effects of steroids on cardiovascular function are not well studied. Early research reported improvements in $\dot{V}O_{2\max}$ with the use of anabolic steroids, possibly as a result of increases in red blood cell production and total blood volume. However, later studies failed to confirm these results, and little research has been done on the effects of steroid administration on the training responses to endurance exercise.

TABLE 16.2 Average Changes in Muscle Size and Strength in Men Taking Testosterone or a Placebo, With and Without Exercise

Variable	Placebo, no exercise	Testosterone, no exercise	Placebo with exercise	Testosterone with exercise
Fat-free mass	+0.8 kg	+3.2 kg	+2 kg	+6.1 kg
Triceps area	-82 mm ²	+424 mm ²	+57 mm ²	+501 mm ²
Quadriceps area	-131 mm ²	+607 mm ²	+534 mm ²	+1,174 mm ²
Bench press strength	No change	+9 kg	+10 kg	+22 kg
Squat strength	+3 kg	+13 kg	+25 kg	+38 kg

Data from Bhasin et al. (1996).

Some athletes use steroids to speed recovery from hard training in the belief that anabolic steroids accelerate recovery and repair of muscle and other tissues. Anecdotal information from professional cyclists and other athletes seems to support this contention. Tamaki and colleagues⁵⁰ reported less muscle fiber damage after a single exhaustive bout of weightlifting in a group of rats that received a single injection of the long-acting androgenic-anabolic steroid nandrolone decanoate compared to a control group that received a placebo. They also found that the rats in the steroid group had an increased rate of protein synthesis during recovery when compared

to the control group. A fuller understanding of the effects of anabolic steroid use on recovery processes will likely develop as more research is conducted. However, regardless of the results, use of anabolic steroids in sport is prohibited.

Risks Associated with Anabolic Steroids

Steroid use in sport is cheating. It is not ethical for athletes to use drugs to improve their chances because doing so violates the very spirit of competition. Most athletes feel that it is wrong for their competitors to artificially improve their performance, yet many of these same athletes recognize that they are unable to compete at the same level if they do not use steroids. Fair competition is not possible if some athletes use illegal substances. This is one of the guiding principles of the World Anti-Doping Code.

The medical risks associated with steroid use are large, especially with the massive doses often used by athletes. And the risks associated with steroid use are not negated by stacking, cycling, and other techniques of steroid use. Children who use steroids before puberty risk early closure of the epiphyses of the long bones, so adult stature is reduced. Use of anabolic steroids suppresses the secretion of gonadotropic hormones, which control the development and function of the gonads (testes and ovaries). In boys and men, decreased gonadotropin secretion can cause testicular atrophy, decreased secretion of testosterone, a reduced sperm count, and impotence long after steroid administration ends. Excess testosterone can lead to higher estrogen production as well, causing enlargement of the male breasts (gynecomastia). In girls and women, gonadotropins are required for ovulation and secretion of estrogens, so a decrease in these hormones disrupts those processes and leads to irregular menstrual cycles. In addition, these hormonal disturbances in women can lead to masculinization: breast regression, enlargement of the clitoris, deepening of the voice, and growth of facial hair. Prostate gland enlargement in men is another possible side effect of steroid use, elevating the risk for prostate cancer.

Many steroids are metabolized by the liver. The tendency for steroids and their metabolites to accumulate in the liver can produce a form of chemical hepatitis and can lead to liver tumors. Abnormal

cardiac hypertrophy (enlarged heart), cardiomyopathy (diseased heart muscle), myocardial infarction (heart attack), thrombosis, arrhythmia, and hypertension have been reported in chronic steroid users. Scientists have found markedly depressed high-density lipoprotein cholesterol (HDL-C) levels—reductions of 30% or more—in athletes on even moderate steroid doses. High-density lipoprotein cholesterol has antiatherogenic properties, meaning that it prevents the development of atherosclerosis. Low levels of HDL-C are associated with a high risk for both coronary artery disease and heart attack (see [chapter 21](#)). Furthermore, low-density lipoprotein cholesterol (LDL-C), which has atherogenic properties, appears to be increased with steroid use.

Substantial personality changes have occurred with steroid use, changes that can be enhanced by abuse of alcohol and other drugs. The most notable change is a marked increase in aggressive behavior, or “roid rage.” Some teens have become extremely violent and have attributed these drastic mood changes to steroid use. Evidence also suggests that drug dependency can result from steroid use.

It is important to note that not everyone using anabolic steroids is an athlete. In fact, it appears that the majority of those taking steroids are nonathletes who use steroids for cosmetic purposes. Further, many self-inject steroids, and when needles are shared, there is a dramatic increase in the risk of contracting infections such as hepatitis and human immunodeficiency virus and acquired immunodeficiency syndrome (HIV/AIDS).

Neither scientists nor physicians know the potential long-term effects of chronic steroid use. One study of male mice that received four different anabolic steroids of the kinds and doses taken by athletes showed that their life span was markedly decreased.¹³ A number of birth defects have been reported by former East German athletes who were unknowingly given steroids during their athletic careers, although the true cause and incidence of these abnormalities remain unknown. Most diseases begin many years before symptoms develop, and it is possible that the more serious health risks of steroid use will not be apparent for 20 or 30 years.

Recent scientific review articles provide more detail on the potential ergogenic effects and health risks associated with anabolic steroid use.^{16,25,31,36,41,60} Most sport governing bodies have developed educational materials for their athletes in hopes of preventing the use of steroids and other PEDs. Also, national governing bodies for most sports have instituted aggressive year-round testing programs in which athletes are tested randomly for steroid use.

Human Growth Hormone

For years, the medical treatment for hypopituitary dwarfism has been administration of **human growth hormone (hGH)**, a hormone secreted by the anterior pituitary gland. Before 1985, this hormone was obtained from cadaver pituitary extracts, and the supply was limited. Since the introduction of genetically engineered hGH in the mid-1980s, availability is no longer an issue.

During the 1980s, realizing this hormone's numerous functions, athletes started using hGH as a possible substitute for, and as a complement to, their use of anabolic steroids. As drug testing for anabolic steroids became more sophisticated, some athletes were looking for an alternative for which there was no drug test at the time.

Proposed Benefits of Growth Hormone

The six normal functions of growth hormone attracted the interest of some coaches and athletes:

- Stimulation of protein and nucleic acid synthesis in skeletal muscle
- Stimulation of bone growth (elongation) if bones are not yet fused
- Stimulation of insulin-like growth factor (IGF-1) synthesis
- Increase in lipolysis, leading to an increase in free fatty acids and an overall decrease in body fat
- Increase in blood glucose levels
- Enhancement of healing after musculoskeletal injuries

Demonstrated Effects of Growth Hormone

Administration of hGH to older men (>60 years) has been shown to increase fat-free mass, decrease fat mass, and increase bone

density.⁴⁵ However, in studies of young men and experienced weightlifters, there appear to be few significant benefits.⁵⁹ More consistently, an increase in collagen synthesis and a reduction in fat mass are associated with hGH, suggesting that hGH is more useful as a cutting (fat-reducing) agent than an anabolic PED.

There is little doubt that exogenous testosterone is a powerful anabolic agent, but hGH does not have the same effects, except in individuals with hGH deficiencies. Exercise and certain amino acid supplements stimulate hGH release from the pituitary, but there is little evidence to suggest that this normal increase in hGH affects muscle protein synthesis or muscle mass and strength. In fact, exercise-related increases in hGH and testosterone are not needed to stimulate muscle protein synthesis and do not influence muscle mass or strength development.⁵⁴

Risks Associated with Growth Hormone Use

As with steroids, potential medical risks are associated with hGH use. Acromegaly can result from taking hGH after the bones have fused. This disorder results in bone thickening, which causes broadening of the hands, feet, and face; skin thickening; and unwanted soft tissue growth. Internal organs typically enlarge. Ultimately, the victim suffers muscle and joint weakness and often heart disease. Cardiomyopathy is the most common cause of death with hGH use. Glucose intolerance, diabetes, and hypertension also can result from hGH use. Athletes who use PEDs often ingest or inject multiple substances, theoretically increasing the risks of adverse health effects.

In Review

- Stimulants such as amphetamines can improve performance in certain sports or activities, but, in addition to being illegal, these drugs carry risks that far outweigh their benefits. Amphetamines can be addictive, and they may mask important afferent and efferent signals that are designed to prevent injury and overexertion.
- Anabolic steroids are more appropriately termed androgenic-anabolic steroids or androgens, because in their natural state they include both androgenic (masculinizing) and anabolic (building) properties. Synthetic steroids have been designed to maximize the anabolic effects while minimizing androgenic effects.

- Anabolic steroid use increases muscle mass and strength and reduces body fat, which can improve sport performance. Aerobic endurance appears to be unaffected by steroid use, but additional research is needed to confirm or contradict that conclusion. Steroid use is illegal in sport and is banned by all governing bodies. Furthermore, the health risks can be considerable.
 - Potential risks associated with use of anabolic steroids include personality changes, “roid rage,” testicular atrophy in men, reduced sperm count in men, breast enlargement in men, breast regression in women, prostate gland enlargement in men, masculinization in women, menstrual cycle disruption in women, liver damage, and cardiovascular diseases.
 - Growth hormone does not appear to have strong anabolic or ergogenic effects. Research supports the ability of hGH to increase fat-free mass and decrease fat mass in older men, but hGH appears to have little or no effect on increasing muscle mass and strength in younger people. Most of the increase in mass and fat-free mass is associated with increased water retention.
 - It appears that hGH has no anabolic properties in young, healthy athletes. However, major health risks are associated with the use of growth hormone. Risks associated with hGH use include acromegaly, hypertrophy of internal organs, muscle and joint weakness, diabetes, hypertension, and heart disease.
-

Diuretics and Other Masking Agents

Diuretics such as desmopressin and acetazolamide affect the kidneys, increasing the loss of electrolytes and therefore water in the urine. Used under the supervision of a physician, diuretics reduce blood volume and total body water. They are generally prescribed to control hypertension and reduce edema (water retention) associated with congestive heart failure or other conditions.

Proposed Benefits of Diuretic Use

Diuretic use by athletes is often for weight control in weight-class sports such as wrestling and among athletes like gymnasts who believe that a lighter body will enhance their performance. Athletes who knowingly consume banned substances often use diuretics to increase their urine volume in the hope that the practice will flush the banned substance out of their bodies and dilute its concentration in the urine, making it more difficult to detect. Other masking agents such as glycerol, dextran, and albumin have been used in an attempt

to expand blood volume and dilute the concentration of banned substances. All are on WADA's list of prohibited substances and techniques.

Demonstrated Effects of Diuretic Use

Diuretics lead to significant temporary weight loss, but no evidence suggests any other potential ergogenic effects. In fact, several side effects make diuretics ergolytic. The fluid loss results primarily from losses in extracellular fluid, including plasma, lowering blood volume. For athletes, particularly those who depend on moderate to high levels of aerobic endurance, this reduction in plasma volume reduces maximal cardiac output, which in turn reduces oxygen delivery and aerobic capacity, impairing performance. In other words, diuretics ensure dehydration.

Risks Associated with Diuretic Use

In addition to reducing plasma volume, the dehydrating effects of diuretics may hinder thermoregulation. As heat production increases during exercise, blood is diverted to the skin so that heat can be lost to the environment. However, when blood plasma volume is diminished, as with diuretic use, skin blood flow is reduced so that blood remains in the central regions of the body to maintain cardiac filling pressure and adequate blood supply and blood pressure to vital organs. Thus, less blood is available to be shunted to the skin, and heat loss may be impaired.

Electrolyte imbalance also can occur with prolonged use of diuretics. Many diuretics cause fluid loss by ensuring electrolyte loss. For example, a diuretic called furosemide inhibits sodium reabsorption in the kidneys, thus allowing more sodium to be excreted in the urine. Because fluid follows the sodium, more fluid is excreted. Electrolyte imbalances can occur with losses of either sodium or potassium. These imbalances can cause fatigue and muscle cramping. More serious imbalances can lead to exhaustion, cardiac arrhythmias, and even cardiac arrest. Some athletes' deaths have been attributed to electrolyte imbalances caused by diuretic use.

β-Blockers

The sympathetic nervous system influences bodily functions through adrenergic nerves: nerves that use norepinephrine as their neurotransmitter. Neural impulses traveling through these nerves trigger the release of norepinephrine, which crosses the synapses and binds to adrenergic receptors on the target cells. These adrenergic receptors are classified into two groups: α -adrenergic receptors and β -adrenergic receptors.

β -Adrenergic blockers, or **β -blockers**, are a class of drugs that block the β -adrenergic receptors, preventing norepinephrine from binding to its receptors on the target cells. Both nonspecific and specific (i.e., cardioselective) forms of β -blockers exist in many different prescription formulations. β -Blockers generally are prescribed for the treatment of hypertension, angina pectoris, and certain cardiac arrhythmias. They also are prescribed as a preventive treatment for migraine headaches, to reduce the symptoms of anxiety and stage fright, and for initial recovery from heart attacks.

Proposed Benefits of β -Blockers

Beta-blocker use by athletes has generally been limited to sports in which anxiety and tremor could impair performance, such as shooting sports, and by golfers seeking to steady their stroke during putting. When a person stands on a force platform (a device that measures mechanical forces), measurable body movement is detected each time the heart beats. These small movements are enough to affect a shooter's aim. Accuracy in shooting sports improves if the rifle or pistol can be shot—or the arrow released—between heartbeats. β -Blockers can slow a shooter's heart rate, allowing more time to stabilize the aim before the next heartbeat. Because of this effect, WADA has banned the use of β -blockers for sports such as archery, billiards, golf, and shooting.

Demonstrated Effects of β -Blockers

Because β -blockers decrease the effects of sympathetic nervous system activity, there is a marked reduction in maximum heart rate. For example, a 20-year-old male athlete with a normal maximum heart rate of 190 beats/min might have a maximum heart rate of only

130 beats/min when taking β -blocking drugs. These drugs also reduce resting heart rate and heart rate during submaximal exercise.

Risks Associated with β -Blockers

Most risks from β -blockers are associated with prolonged use, not isolated use as might happen with athletes. By blocking the sympathetic nervous system's relaxation effect on smooth muscle, β -blockers can induce bronchospasm in people with asthma and can cause cardiac failure in people who have underlying problems with cardiac function. In people with bradycardia, β -blocker medications can lead to heart block. In addition, the decreased blood pressure caused by β -blockers can result in light-headedness. Some people with type 2 diabetes can become hypoglycemic on β -blockers because insulin secretion is no longer limited. For most athletes, β -blockers are ergolytic because of the reduction in maximal heart rate and feelings of pronounced fatigue. For athletes who are prescribed β -blockers for a medical condition such as hypertension or a heart arrhythmia, β -1-selective blockers are usually preferred because they have fewer negative effects on performance.

Blood Doping

Blood doping refers to practices that alter the oxygen-carrying capacity of blood. These include infusions of red blood cells, artificial hemoglobin, or erythropoietin to stimulate the body to produce more red blood cells. Erythropoietin is the naturally occurring hormone that stimulates red blood cell production. It increases the number of red blood cells and therefore the blood's oxygen-carrying capacity.

Blood doping is prohibited because transfusion of red blood cells, previously donated by either the recipient (autologous transfusions) or someone else with the same blood type (homologous transfusions), increases red blood cell mass and the oxygen-carrying capacity of blood. Use of erythropoietin (EPO) or EPO-stimulating substances that increase red blood cell mass is prohibited for the same reason and also because numerous deaths have been associated with EPO use.

Proposed Benefits of Blood Doping

The premise underlying blood doping is simple. Because most oxygen is carried in blood bound to hemoglobin, increasing the number of red blood cells (and hemoglobin) available to ferry the oxygen to muscles could benefit endurance performance.

Demonstrated Effects of Blood Doping

In a landmark study in the early 1970s, Ekblom and associates²⁴ withdrew between 800 and 1,200 ml of blood from their subjects, refrigerated the blood, then reinfused the red blood cells into the subjects about 4 weeks later. Results showed a considerable improvement in $\dot{V}O_{2\max}$ (9%) and treadmill performance time (23%) after reinfusion. Over the next few years, several studies confirmed these original findings, although several others failed to demonstrate an ergogenic effect.

Thus, the research literature was divided on the effectiveness of blood doping until a major breakthrough occurred in 1980 as a result of a study by Buick and colleagues.¹⁵ Eleven highly trained distance runners were tested at different times during the study: (1) before blood withdrawal, (2) following blood withdrawal after allowing adequate time to reestablish normal red blood cell levels but before reinfusion of the removed blood, (3) following a sham reinfusion of 50 ml of saline (a placebo), (4) following reinfusion of 900 ml of the subject's own blood that originally had been withdrawn and preserved by freezing, and (5) after the elevated red blood cell levels had returned to normal.

As shown in figure 16.5, the researchers found a substantial increase in $\dot{V}O_{2\max}$ and treadmill running time to exhaustion after the reinfusion of the red blood cells and no change after the sham reinfusion. This increase in $\dot{V}O_{2\max}$ persisted for up to 16 weeks, although the improvement in treadmill time was lost within the first 7 days.

Why was the Buick study a major breakthrough? Gledhill²⁸ helped explain the controversy arising from the early studies. Many early studies that showed no improvement with blood doping had reinfused only small volumes of red blood cells, and the reinfusion was conducted within 3 to 4 weeks of the blood withdrawal. First, it appears 900 ml or more of whole blood must be reinfused to have an

effect on performance. Increases in $\dot{V}O_{2\max}$ and performance are not as great when smaller volumes are used. In fact, some studies using smaller volumes failed to show any differences in $\dot{V}O_{2\max}$ or performance.

Second, it appears that it is necessary to wait for at least 5 weeks and possibly as long as 10 weeks before reinfusion. This is based on the time it takes to reestablish the blood's original, prewithdrawal red blood cell mass and hematocrit.

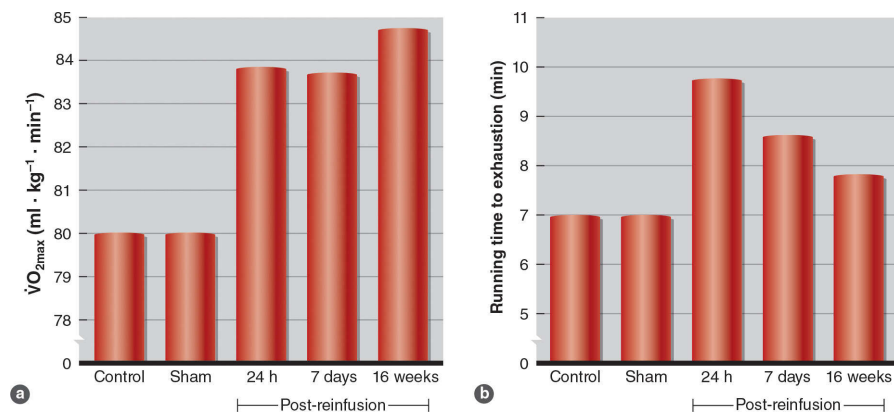


FIGURE 16.5 Changes in (a) $\dot{V}O_{2\max}$ and (b) running time to exhaustion following reinfusion of red blood cells.

(a) Based on Buick et al. (1980). (b) Adapted by permission from F.J. Buick et al., "Effect of Induced Erythocytomia on Aerobic Work Capacity," *Journal of Applied Physiology* 48 (1980): 636-642.

Finally, researchers who conducted early studies refrigerated the withdrawn blood. When blood is refrigerated, approximately 40% of the red blood cells are destroyed; maximal storage time for blood under refrigeration is approximately 5 weeks. Later studies used frozen storage. Freezing allows an almost unlimited storage time, and only about 15% of the red blood cells are lost. Gledhill²⁸ concluded that blood doping significantly improves $\dot{V}O_{2\max}$ and endurance performance when done under optimal conditions:

- A minimum of 900 ml of blood reinfusion
- A 5- to 6-week minimum interval between withdrawal and reinfusion
- Blood storage by freezing

The researchers also showed that the improvements in $\dot{V}O_{2\max}$ and performance are the direct result of the blood's increased hemoglobin content, not an increased cardiac output caused by an expanded plasma volume.

Does an increase in $\dot{V}O_{2\max}$ and treadmill time as a result of blood doping translate into improved endurance performance? Several studies have addressed this issue. Researchers in one study observed 5 mi (8 km) treadmill run times in a group of 12 experienced distance runners.⁵⁹ Their times were checked before and after saline (placebo) infusion and before and after blood infusion; the run times on the treadmill were significantly faster following blood infusion, but this difference became clear only over the second half of the trial. The blood infusion trials were 33 s (3.7%) faster over the last 2.5 mi (4 km) and 51 s (2.7%) faster over the full distance, compared to the placebo trials. A second study looked at 3 mi (4.8 km) run times in a group of six trained distance runners and reported an average decrease of 23.7 s following blood doping when compared to the runners' blind control trials.²⁹ Subsequent studies confirmed improvements in distance running and cross-country skiing performance with blood doping.^{23,48} Figure 16.6 illustrates the improvement in run time with blood doping for distances of up to 11 km (6.8 mi).

Risks Associated with Blood Doping

Although blood doping is relatively safe in the hands of competent physicians, it has inherent dangers.³ Adding more red blood cells to the cardiovascular system can cause the blood to become too viscous, which can lead to clotting and possibly heart failure. In an effort to control blood doping, some sport governing bodies, such as professional cycling, do not allow athletes to compete if their hemoglobin concentrations or hematocrit levels are too high (e.g., >50%). With autologous blood transfusions, in which the recipients receive their own blood, mislabeling of the blood can occur. With homologous blood transfusions, in which blood is received from a matched donor, several other complications can occur. The reinfused blood could be mismatched and an allergic reaction could be

triggered. The athlete also risks infection from hepatitis or HIV pathogens.⁵¹

The outcome of EPO use is less predictable than that of red blood cell reinfusion (blood doping). Once EPO has been administered, it is difficult to predict how much red blood cell production will occur. This places the athlete at great risk of substantial increases in blood viscosity and associated problems such as thrombosis (blood clot), myocardial infarction (heart attack), congestive heart failure, hypertension, stroke, and pulmonary embolism.

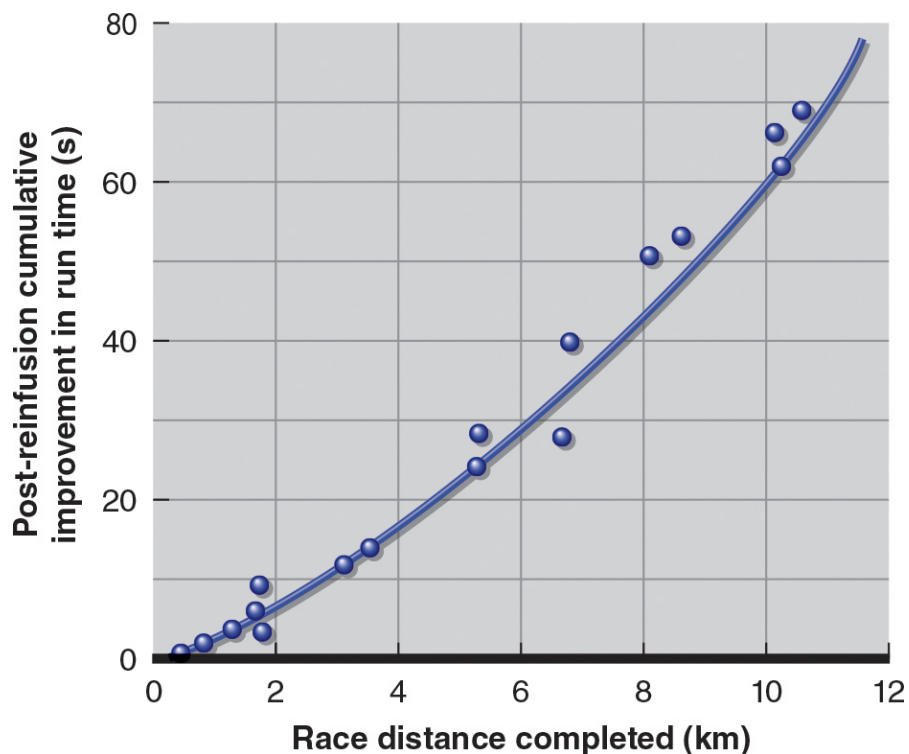


FIGURE 16.6 Improvements in running times for distances of up to 11 km (6.8 mi) following reinfusion of red blood cells from two units of blood preserved by freezing. Values on the y-axis reflect the reduction in time to run a specific distance on the x-axis. As an example, for a 10 km (6.2 mi) race, one could expect to run 60 s faster after reinfusion.

Adapted by permission from L.L. Spriet, Blood Doping and Oxygen Transport, in *Ergogenics: Enhancement of Performance in Exercise and Sport*, edited by D.R. Lamb and M.H. Williams (Dubuque, IA: Brown & Benchmark, 1991), 213-242. Copyright 1991 Cooper Publishing Group, Carmel, IN.

The potential risks of blood doping, even without consideration of the legal and ethical issues involved, outweigh any potential benefits.

In Review

- Diuretics affect the kidneys, increasing urine production. They often are used by athletes for temporary weight reduction and also by those trying to mask the use of other drugs in anticipation of drug testing.
- Weight loss (dehydration) is the only substantiated potential ergogenic effect of diuretics, but this weight loss is primarily from the extracellular fluid compartment, including blood plasma. This can lead to dehydration, volume depletion, increased cardiac strain, and electrolyte imbalances.
- β -blocker medications are banned in sports in which a slowed heart rate can provide a competitive advantage. The World Anti-Doping Agency maintains a list of specific sports in which β -blockers are prohibited substances.
- β -blockers block β -adrenergic receptors, limiting the binding of catecholamines. They slow the resting heart rate, which is a distinct advantage for shooters who try to release the arrow or squeeze the trigger between heartbeats to minimize the slight tremor associated with each beat, and could be an advantage for golfers when chipping and putting.
- β -blockers can cause heart block, hypotension, bronchospasm, pronounced fatigue, and decreased motivation. Selective β -blockers have fewer side effects than nonselective blockers.
- Blood doping and administration of EPO can enhance aerobic capacity and the performance of aerobic sports or activities. This occurs through an increase in the oxygen-carrying capacity of blood, primarily attributable to increased red blood cells. Both procedures involve risk.
- Studies have shown major increases in maximal oxygen uptake, time to exhaustion, and actual performance in cross-country skiing, cycling, and distance running as a result of blood doping.
- Erythropoietin is the naturally occurring hormone that stimulates red blood cell production in bone marrow. Studies have clearly demonstrated increased maximal oxygen consumption and increased exercise time to exhaustion after administration of EPO.
- Serious risks associated with blood doping include blood clotting, heart failure, and, if blood from another donor is used by accident or intentionally, transfusion reactions and transmission of hepatitis and HIV.
- Because the magnitude of the response to EPO administration cannot be accurately predicted, EPO use can be dangerous. The hormone can lead to death if red blood cells are overproduced, increasing the blood's viscosity. Known risks

include thrombosis, myocardial infarction, congestive heart failure, hypertension, stroke, and pulmonary embolism.

IN CLOSING

All forms of ergogenic aids are associated with potential benefits and possible risks. That risk-to-benefit ratio is greater for some interventions than for others. For example, there are potential benefits to creatine loading for some athletes, but some risk is involved if the supplement contains banned substances. In addition, athletes must recognize the legal, ethical, and medical consequences of using any ergogenic agent, especially those that are expressly prohibited (e.g., steroid use, blood doping) or that may contain banned substances (e.g., contaminated substances.) Athletes who use banned substances or procedures risk disqualification from a particular competition and can be banned from competition in their sport for a year or more. We have discussed pharmacological, hormonal, physiological, and some specific ergogenic nutrition aids. In the next part of the book, we shift our focus to younger, older, and female athletes within the broader categories of growth and development, aging, and sex differences in exercise performance. We begin in [chapter 17](#) by examining special considerations in the child and adolescent.

KEY TERMS

amphetamines
anabolic steroids
 β -blockers
blood doping
caffeine
creatine
diuretics
ephedrine
ergogenic
ergolytic
human growth hormone (hGH)
placebo effect
pseudoephedrine
testosterone
therapeutic use exemption

STUDY QUESTIONS

1. What is the meaning of the term *ergogenic aid*? What is an ergolytic effect?
2. Why is it important to include control groups and placebos in studies of the ergogenic properties of any substance or phenomenon?

3. How do β -alanine and sodium bicarbonate function as possible ergogenic aids?
4. How might caffeine improve athletic performance?
5. How might the polyphenols in cherry juice benefit recovery from intense exercise?
6. What are the potential ergogenic properties of creatine supplementation?
7. What roles do the amino acid leucine play in building muscle mass?
8. What effect does nitrate ingestion have on the response to exercise?
9. By what criteria are substances considered for the banned list, according to the World Anti-Doping Code?
10. What is presently known about the use of amphetamines in athletic competition? What are the potential risks of using amphetamines?
11. What are the effects of anabolic steroid use on athletic performance? What are some of the medical risks of steroid use?
12. What is known about hGH as a potential ergogenic aid? What are the risks associated with its use?
13. Can diuretics be ergogenic? What are some risks associated with their use?
14. Under what circumstances might β -blockers be ergogenic aids?
15. What is blood doping? Does blood doping improve athletic performance?
16. By what mechanism is erythropoietin theorized to benefit performance?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.